Catherine Yeh

CS PhD Candidate · Harvard University

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Education

Harvard University — *Boston, MA*
PH.D. IN COMPUTER SCIENCE WITH MASTERS IN PASSING — *2022 - present*
Advisors: Martin Wattenberg, Fernanda Viégas, Hanspeter Pfister
Research Interests: Data Visualization, Human-AI Interaction, Interpretability

Williams College — *Williamstown, MA*
B.A. IN COMPUTER SCIENCE & COGNITIVE SCIENCE — *2018 - 2022*
Advisor: Iris Howley – Thesis: Toward an Empirical Framework for Post-hoc Explainable AI
Graduated Summa Cum Laude (top 2% of class) with Highest Honors in Computer Science

Research Experience

Industry

Google DeepMind – Inception Team — *Mountain View, CA*
STUDENT RESEARCHER — *Summer 2026*
Mentors: Peggy Chi, Lei Shi
Developed new interaction techniques for expressing and steering design intent when vibe coding.

Adobe Research – STORIE Lab — *Seattle, WA*
RESEARCH SCIENTIST INTERN — *Summer 2025*
Mentors: Bryan Wang, Anh Truong, Mira Dontcheva
Designed a video authoring tool to help creators craft compelling video stories by blending captured and generative media (C6).

Apple Research – Human-Centered Machine Intelligence Group — *Seattle, WA*
RESEARCH INTERN — *Summer 2024*
Mentors: Fred Hohman, Donghao Ren, Yannick Assogba, Dominik Moritz
Developed a visual, human-in-the-loop data augmentation tool for increasing data diversity in text datasets (C3).

Microsoft Research – Human Understanding & Empathy Group — *Redmond, WA*
RESEARCH INTERN — *Summer 2023*
Mentors: Gonzalo Ramos, Rachel Ng
Designed a personalized environment to preserve agency and ownership while writing with large language models (P1).

Adobe Research – Media Intelligence Lab — *San Jose, CA*
RESEARCH SCIENTIST INTERN — *Summer 2022*
Mentors: Franck Dernoncourt, Nedim Lipka
Developed prototypes showcasing vision for the next-gen document reader (W2).

Microsoft Research – Productivity & Intelligence Group — *Redmond, WA*
RESEARCH INTERN — *Summer 2021*
Mentors: Jenna Butler, Christian Bird
Built a self-reflection chatbot and personalized dashboards for studying hybrid workforce productivity and well-being (O1).

Academic

Harvard University – Insight & Interaction Lab / Visual Computing Group — *Boston, MA*
GRADUATE RESEARCH ASSISTANT — *2022 - present*
Advisors: Martin Wattenberg, Fernanda Viégas, Hanspeter Pfister
Designing visualization-driven interfaces and techniques to support model interpretability and enable more productive, safe, and creative human-AI interactions (C5, C4, C2, W3).

Williams College – Computation^2 Lab *Williamstown, MA*

Undergraduate Research Assistant *2021-2022*

Advisor: Molly Feldman

Studied and developed guidelines for replicability in human-computer interaction research.

Williams College – Human-AI Interaction Lab *Williamstown, MA*

Undergraduate Research Assistant *2019-2022*

Advisor: Iris Howley

Studied knowledge tracing algorithms and developed a novel, evidence-based framework for explainable AI (C1, W1).

Williams College – Concepts & Categories Lab *Williamstown, MA*

Undergraduate Research Assistant *2018-2019*

Advisor: Safa Zaki

Designed eye-tracking experiments to study cognitive learning strategies in categorization tasks.

Publications

Conference

** denotes equal contribution*

C6 **Vidmento: Creating Video Stories Through Context-Aware Expansion With Generative Video**
Catherine Yeh, Anh Truong, Mira Dontcheva, Bryan Wang
ACM Conference on Human Factors in Computing Systems (CHI), 2026.
Paper

C5 **Story Ribbons: Reimagining Storyline Visualizations With Large Language Models**
Catherine Yeh, Tara Menon, Robin Singh Arya, Helen He, Moira Weigel, Fernanda Viégas, Martin Wattenberg
IEEE Visualization Conference (VIS), 2025.
Paper · 🏆 Best Paper Honorable Mention

C4 **Chronotome: Real-Time Topic Modeling for Streaming Embedding Spaces**
Matte Lim*, Catherine Yeh*, Martin Wattenberg, Fernanda Viégas, Panagiotis Michalatos
IEEE Visualization Conference (VIS), 2025.
Paper · 🏆 Best Paper Honorable Mention

C3 **Exploring Empty Spaces: Human-in-the-Loop Data Augmentation**
Catherine Yeh, Donghao Ren, Yannick Assogba, Dominik Moritz, Fred Hohman
ACM Conference on Human Factors in Computing Systems (CHI), 2025.
Paper · 🏆 Best Paper Honorable Mention

C2 **AttentionViz: A Global View of Transformer Attention**
Catherine Yeh, Yida Chen, Aoyu Wu, Cynthia Chen, Fernanda Viégas, Martin Wattenberg
IEEE Visualization Conference (VIS), 2023.
Paper

C1 **Designing for Student Understanding of Learning Analytics Algorithms**
Catherine Yeh, Noah Cowit, Iris Howley
International Conference on Artificial Intelligence in Education (AIED), 2023.
Paper

Workshop

W3 **What Kind of User Are You? Uncovering User Models in LLM Chatbots**
Yida Chen, Aoyu Wu, Trevor DePodesta, Catherine Yeh, Lena Armstrong, Kenneth Li, Nicholas Castillo Marin, Oam Patel, Jan Riecke, Shivam Raval, Olivia Seow, Martin Wattenberg, Fernanda Viégas
ICML Actionable Interpretability Workshop (AIW), 2025
Paper

W2 **Envisioning the Next-Gen Document Reader**
Catherine Yeh, Franck Dernoncourt, Nedim Lipka
AAAI Workshop on Scientific Document Understanding (SDU), 2023.
Paper

W1 **Tests, Memory, and Artificial Intelligence: How Can We Know What People Know?**
Noah Cowit, Catherine Yeh, Iris Howley
IEEE VIS Workshop on Visualization for AI Explainability (VISxAI), 2019.

Other

O1 **Walk a Mile in Their Shoes: The Covid Pandemic Through the Lens of Four Tech Workers**
Jenna Butler, Catherine Yeh
Communications of the ACM & ACM Queue, 2022.
Paper · Paper 2

Preprint / In Submission

P3 **VideoTrail: Interactive Video Foraging Through Hierarchical Visualization**
Catherine Yeh, Bryan Wang, Fernanda Viégas, Martin Wattenberg

P2 **Neural Representations As New Modes Of Interactivity**
Stanley Huang, Anthony Baez, Catherine Yeh, Pat Pataranutaporn

P1 **GhostWriter: Augmenting Collaborative Human-AI Writing Experiences Through Personalization and Agency**
Catherine Yeh, Gonzalo Ramos, Rachel Ng, Andy Huntington, Richard Banks
Paper

Patents

Video Sequencing and Gap Detection for Authoring Videos Including Captured and Generated Content
Application Filed

Generative Neural Network Model Style Guide Management, US Patent App. 18400715
August 2025

Invited Talks & Panels

Crash Course on Securing PhD Research Internships, Harvard University Computer Science Department
November 2025

Principles of Data Visualization, Carnegie Mellon University Language Technologies Institute
June 2025

Exploring Empty Spaces: Human-in-the-Loop Data Augmentation, Apple Research Machine Learning Summit
March 2025

Panel on Effective Research Practices for PhD Students, Harvard University Computer Science Department
February 2025

Honors & Awards

2025 Best Paper Honorable Mention (x2), IEEE VIS
Best Paper Honorable Mention, ACM CHI
Graduate Collegiate Award Finalist, NCWIT
10 Under 10 Outstanding Alumni Award Recipient, The Overlake School

2024 Graduate Research Fellowship Recipient, NSF
Graduate Fellowship Recipient, Kempner Institute

2022 Inductee, Sigma Xi International Scientific Research Honor Society
Outstanding Undergraduate Researcher Award Runner Up, CRA

2021 Undergraduate Collegiate Award Finalist, NCWIT
Women-in-Technology Scholarship Finalist, Adobe Research
Junior Year Inductee (top 5% of class), Phi Beta Kappa National Honor Society
ACM Student Research Competition Finalist, Grace Hopper Conference
Ward Prize Finalist, Williams College

2019 **Computer Science Class of 1960s Scholar (2019 - 2022)**, Williams College
3rd Place Winner, National USCLAP Statistics Competition
Summer Science Research Fellowship Recipient, Williams College
Scholarship Recipient, Grace Hopper Conference

2018 **Dean's List (2018 - 2022)**, Williams College

Other Industry Experience

Oracle Cloud Infrastructure (OCI) *Seattle, WA*
Software Engineer Intern *Summer 2020*
Built dashboard and visual database for regional compliance programs.

Sunshine *Palo Alto, CA*
Software Engineer Intern *Winter 2020*
Created dashboard for AI-driven consumer app.

Teaching Experience

CS279R: Research Topics in Human-Computer Interaction, Harvard University
Fall 2023

CSCI 334: Principles of Programming Languages, Williams College
Spring 2022, Fall 2021, Spring 2021

CSCI 315: Computational Biology, Williams College
Fall 2021

CSCI 256: Algorithm Design & Analysis, Williams College
Spring 2020

CSCI 136: Data Structures & Advanced Programming, Williams College
Fall 2020, Fall 2019, Spring 2019

Mentoring

Research

Allison Huang, University of Southern California '26
Summer 2026 – Generative UIs for Verifying LLM Reasoning

Helen He, Harvard University '26
Spring 2025 – LLM-Augmented Storyline Visualizations

Vicki Xu, Harvard University '23
Fall 2023 - Logit Lens on Vision Transformer

Cynthia Chen, Harvard University '24
Spring 2023 – Visualizations for Transformer Interpretability

Other

VISBuddies Mentor, IEEE VIS
2025 - present

SEAS Research Program Mentor, Harvard University
2024 - present

Women in STEM Mentor, Harvard University
2022 - present

Underrepresented Identities in CS Mentor, Williams College
2020 - 2022

Service

Organizer

IEEE VIS Workshop on Visualization for AI Explainability (VISxAI) – 2025, 2026
IEEE VIS Workshop on GenAI, Agents, and the Future of VIS (VISxGenAI) – 2025

Program Committee

IEEE VIS Workshop on GenAI, Agents, and the Future of VIS (VISxGenAI) – 2026
IEEE VIS Workshop on Visualization for Communication (VisComm) – 2025
IEEE VIS Workshop on Visualization for AI Explainability (VISxAI) – 2024

Reviewer ** indicates outstanding review recognition*

Conference

ACM Symposium on User Interface Software and Technology (UIST) – 2026*
IEEE Eurographics Conference on Visualization (EuroVis) – 2026*
IEEE Pacific Visualization Conference (PacificVis) – 2026
IEEE Visualization Conference (VIS) – 2025
ACM Conference on Human Factors in Computing Systems (CHI) – 2025*, 2026*, 2027
ACM Conference on Designing Interactive Systems (DIS) – 2024

Journal

Journal of Visualization and Interaction (JoVI) – 2025
ACM Transactions on Intelligent Systems and Technology (TIST) – 2024

Volunteer

IEEE VIS Student Volunteer – 2023

Misc

Art Portfolio